

Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3445U10-1



THURSDAY, 13 JUNE 2024 – MORNING

APPLIED SCIENCE (Double Award)
UNIT 1: Energy, Resources and the Environment
FOUNDATION TIER
1 hour 30 minutes

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **3** is a quality of extended response (QER) question where your writing skills will be assessed.

You are reminded to show all your workings. Credit is given for correct workings even when the final answer given is incorrect.

A Periodic Table is printed on page 20.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	15	
3.	6	
4.	12	
5.	12	
6.	19	
Total	75	

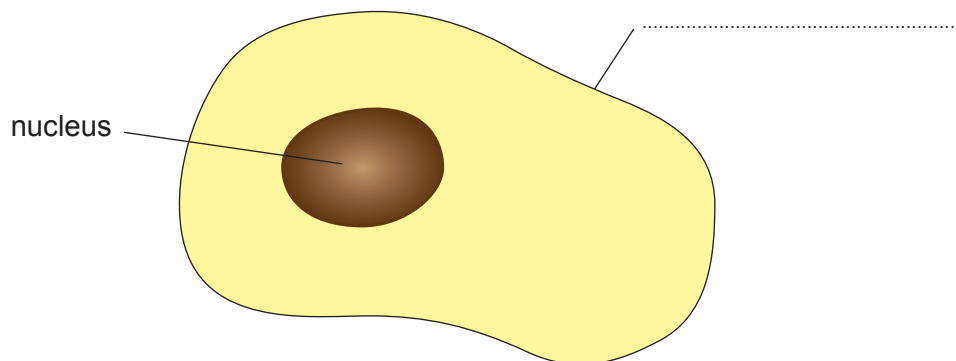


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Answer **all** questions.

1. The diagram below shows an animal cell.

chloroplast	vacuole	cell membrane
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(a) (i) Use the correct word(s) from the box to complete the labelling of the animal cell. [1]

(ii) Tick (✓) the box next to a structure found inside the nucleus. [1]

chloroplast ☐

chromosome ☐

cellulose ☐

(iii) Tick (✓) the box next to the correct statement about the function of the nucleus. [1]

gives structural support of the cell ☐

controls the activity of the cell ☐

the site of photosynthesis ☐



- (b) Cells are organised into tissues. Complete the following sentences by underlining the correct word in each bracket. [2]

Tissues are a collection of many (**different** / **similar** / **bigger**) cells.

Tissues work together to perform a (**smaller** / **particular** / **bigger**) function.

- (c) The cell membrane controls the entry and exit of all substances into and out of the cell. Many substances are transported into the cell by diffusion.

- (i) Complete the following sentence by underlining the correct word in each bracket. [2]

Diffusion is the movement of substances from a region of (**high** / **low** / **zero**) concentration to a region of (**higher** / **lower** / **negative**) concentration.

- (ii) Oxygen is a gas that diffuses into the cell. Tick (✓) the box next to the gas that diffuses **out of** the cell. [1]

argon

☐

carbon dioxide

☐

helium

☐

- (iii) I. The oxygen that diffuses into the blood is needed by all cells to release energy.

Circle the process that releases energy. [1]

excretion

respiration

absorption

- II. Complete the equation for this process using the correct terms from the box. [2]

glucose

protein

sulfur

salt

carbon dioxide

..... + oxygen → + water



2. This question is about acids and alkalis. Lisa started filling in the following table.

Chemical name	Chemical formula	Colour with Universal Indicator	pH	Acid, alkali or neutral
sulfuric acid	H ₂ SO ₄	red	1	acid
hydrochloric acid	HCl	red		acid
calcium hydroxide	Ca(OH) ₂	purple	12	alkali
sodium hydroxide	NaOH		14	alkali
water		green	7	

(a) Complete the table.

[4]

(b) Lisa added magnesium to sulfuric acid. She recorded the temperature of the mixture during the reaction.

The word equation for this reaction is shown below.

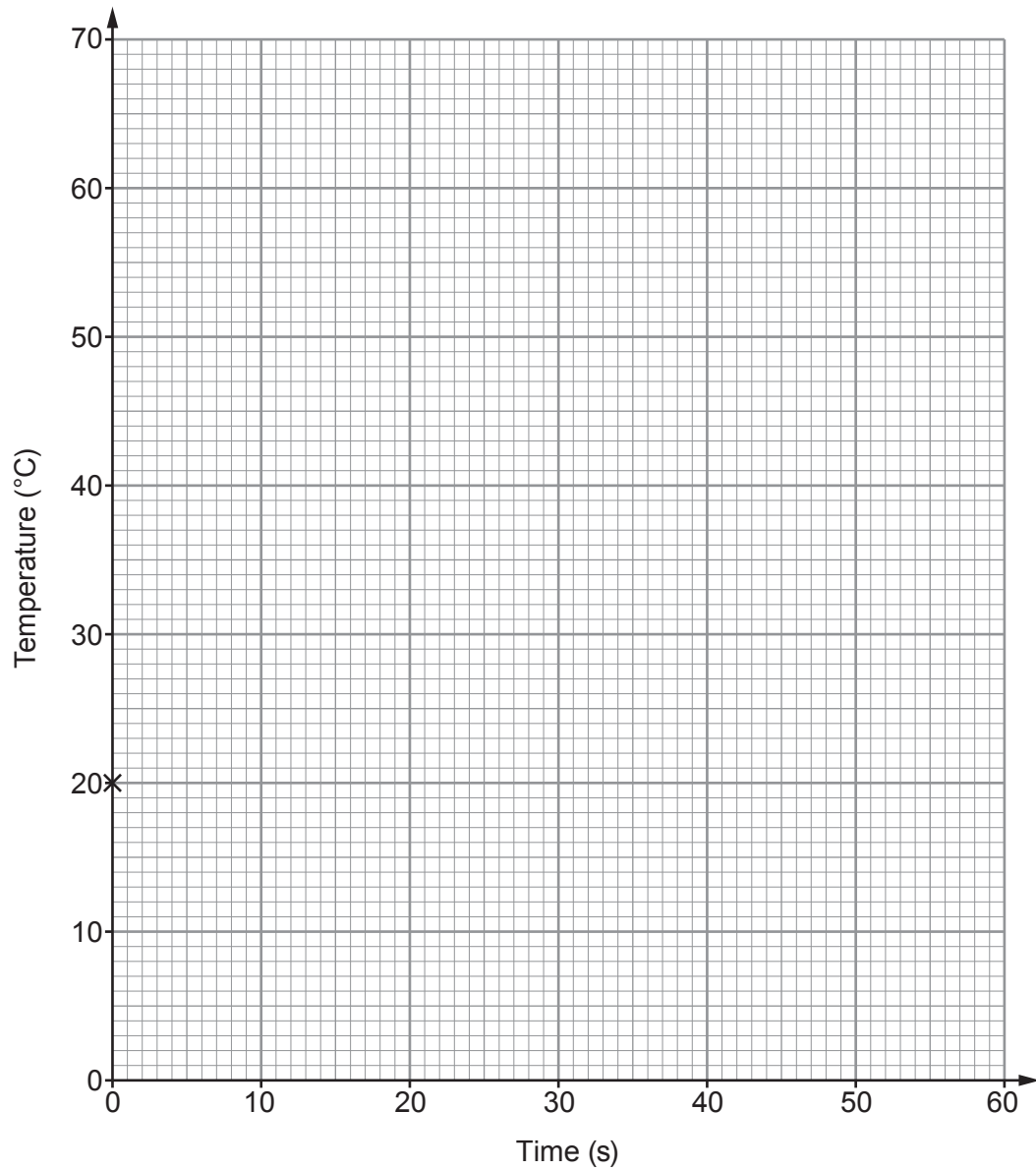


The temperatures are shown in the table below.

Time (s)	Temperature (°C)
0	20
10	40
20	50
30	60
40	60
50	58



- (i) Use the data in the table to plot a graph on the grid below and join the points with a ruler. The first point has been plotted for you. [3]



- (ii) Calculate the temperature change during the first 20 seconds. [2]

temperature change = °C



(iii) Complete the following table to describe how the temperature changes over the 50 seconds. The first two rows have been completed for you. [3]

Time (s)	Temperature decreases	Temperature stays the same	Temperature increases
0–10			✓
10–20			✓
20–30			
30–40			
40–50			

(c) Sulfuric acid has the formula H_2SO_4 .

Complete the following table to calculate the relative formula mass of H_2SO_4 . [3]

Element	Number of atoms	Relative atomic mass (A_r)	Total mass of the element in H_2SO_4
hydrogen	2	1	
sulfur	1	32	
oxygen	4	16	64
relative formula mass (M_r)			



- Describe the role of the enzyme amylase (carbohydhrase) in the digestive system. [6 QER]

- where amylase is made
- how amylase works
- the conditions necessary for its action

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6



4. (a) Welsh Water is responsible for supplying and maintaining safe water supplies in Wales. It monitors the composition of our drinking water.

The table below shows the composition of stream water supplying a reservoir in Wales. The samples were taken from three different streams.

Component	Concentration (units)		
	stream 1	stream 2	stream 3
sodium ions (Na^+)	5.0	5.0	5.0
potassium ions (K^+)	6.3	6.2	6.2
magnesium ions (Mg^{2+})	15.3	15.4	15.5
calcium ions (Ca^{2+})	50.4	50.1	50.1
chloride ions (Cl^-)	30.1	30.0	30.2
dissolved sulfur dioxide	6.0	8.2	10.0
dissolved carbon dioxide	0.6	0.9	1.6
pH	5.7	5.5	5.4

Use the information in the table to answer the following questions.

- (i) State the name of **one** ion whose concentration is the same in each stream. [1]

.....

- (ii) There is more dissolved sulfur dioxide in stream 3 than in stream 1.

I. Use the equation:

$$\% \text{ difference} = \frac{\text{concentration in stream 3} - \text{concentration in stream 1}}{\text{concentration in stream 3}} \times 100$$

to calculate the percentage difference between the dissolved sulfur dioxide in stream 1 and stream 3. [2]

percentage difference =



- II. Complete the following sentence by underlining the correct term in the bracket.

An increase in dissolved sulfur dioxide makes the rainwater (**more acidic** / **become neutral** / **less acidic**). [1]

- III. State how an increase in dissolved sulfur dioxide affects the pH of water. [1]

.....

- (iii) A technician believes that streams 1, 2 and 3 have water of the same hardness. Use the data in the table to explain whether you agree with the technician. [3]

.....

.....

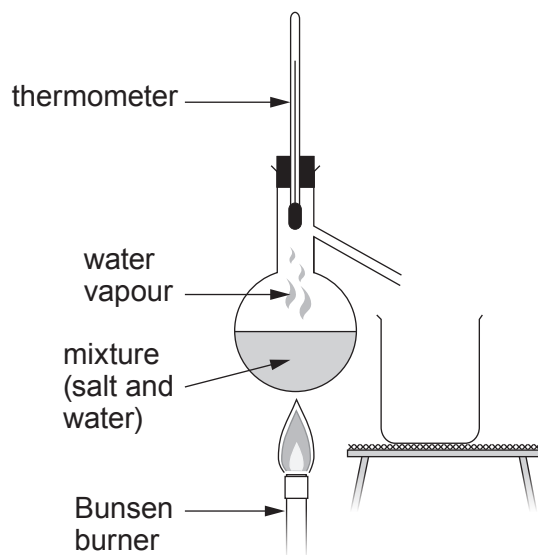
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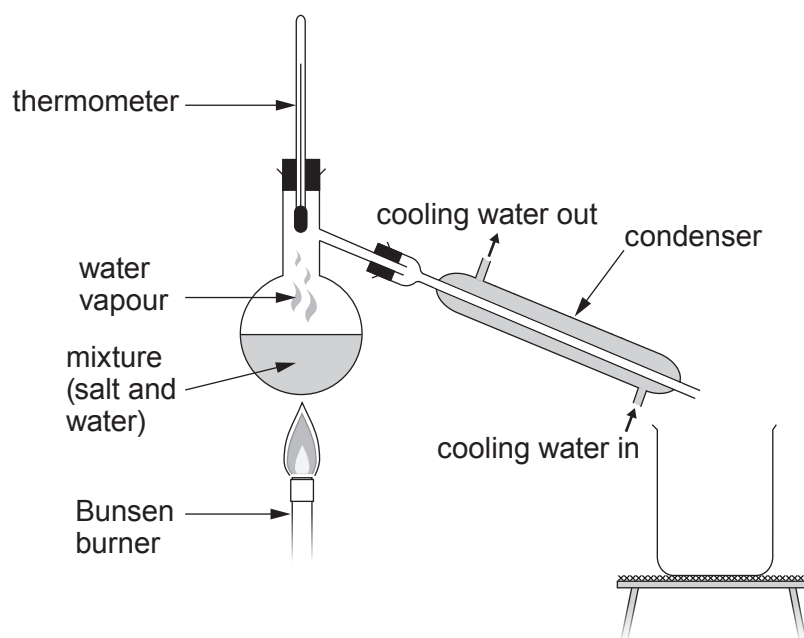
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- (b) When water is in short supply, seawater can be desalinated. In a school laboratory, distillation apparatus can be used to desalinate seawater. A student was given the choice of apparatus A or B.



Apparatus A



Apparatus B



- (i) State the temperature that the thermometer will record during the desalination of water. [1]

..... °C

- (ii) A student believes that she will collect more water using apparatus A than apparatus B.

Explain whether you agree. [2]

.....

.....

.....

- (iii) South East England needs an alternative source of water to supply drinking water. One of the options is to build a desalination plant.

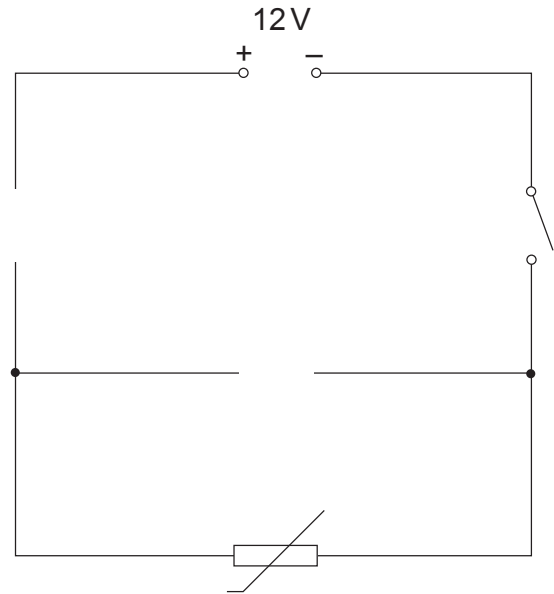
State **one** disadvantage of desalination of sea water. [1]

.....

.....



5. Tommy and Arthur are investigating the effect of temperature on the resistance of a thermistor. The circuit diagram drawn by Tommy is shown below.



- (a) Complete the circuit above by adding:
- (i) an **ammeter** to measure current through the thermistor. [1]
 - (ii) a **voltmeter** to measure voltage across the thermistor. [1]
- (b) The results of Tommy’s experiment are shown in the table below.

Temperature (°C)	Current (A)	Voltage (V)	Resistance (Ω)
0	0.05	12	240
20	0.10	12	120
40	0.20	12	60
60	0.40	12	30
80	0.80	12	15
90	1.1	12	

Use only the information in the table to answer the following questions.

- (i) State the independent variable in this experiment. [1]
-
- (ii) State a controlled variable in this experiment. [1]
-



(c) Use the information in the table to answer the following questions.

(i) Estimate the **current** when the temperature is 100 °C.

[2]

current = A

(ii) Use the equation:

$$\text{resistance} = \frac{\text{voltage}}{\text{current}}$$

to calculate the resistance of the thermistor at 90 °C.

[2]

resistance = Ω

(iii) Tommy thinks that when he increases the temperature by 20 °C the resistance of the thermistor decreases by half every time. Use Tommy's results to explain whether you agree.

[2]

.....

.....

.....

(iv) Arthur wants to know if the thermistor they used in the circuit would be suitable as a temperature sensor. The sensor must work between 20 °C to 60 °C and vary in resistance by at least 100 Ω .

Use Tommy's results to explain whether the thermistor would be suitable.

[2]

.....

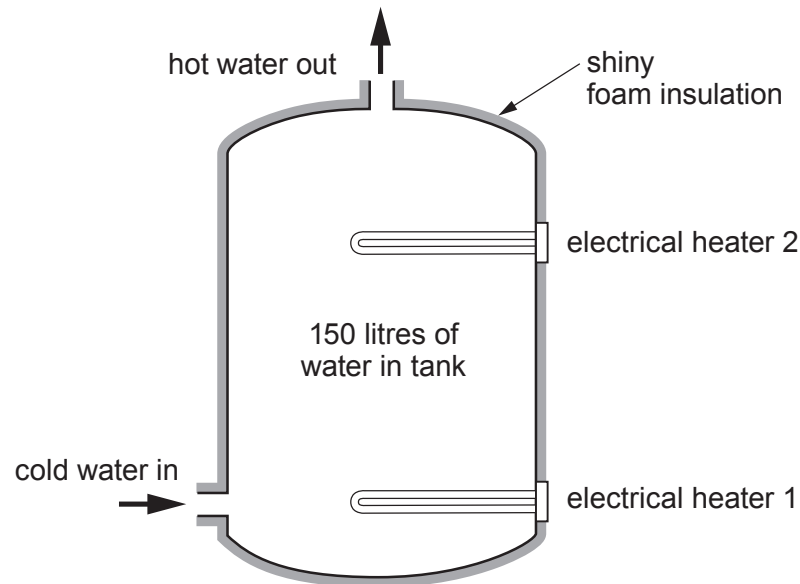
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6. Paul lives in an apartment in central London. His electricity bill is £80 a month. Paul wants to replace his old electric boiler with a new one. A local plumbing company has suggested using a new boiler, the Redstar boiler, which has two electrical heaters. Paul will be able to use either heater to provide hot water after the boiler is installed.

Information about the Redstar boiler



Savings

Cost of boiler	£2500.00
Typical savings	£15.00 per month

Technical information on the two heaters in the Redstar boiler

Information	heater 1	heater 2
maximum volume of water that is heated by heater (litres)	150	50
time to heat this volume of water (hours)	3	0.5
total power supplied (kW)	2	4
power used to heat the water (kW)	1.6	3.6
efficiency (%)	80	
cost to heat the water (p)		44

- (a) Use the information given about the boiler to answer the following questions.

- (i) Calculate the expected payback time for the boiler in years.

[3]

payback time = years



(ii) Use the equation:

$$\% \text{ efficiency} = \frac{\text{power usefully transferred}}{\text{total power supplied}} \times 100$$

to calculate the efficiency of **heater 2**.

[2]

% efficiency =

(iii) Use the equations:

$$\text{units used (kWh)} = \text{power (kW)} \times \text{time (h)}$$

$$\text{total cost} = \text{cost of one unit} \times \text{units used}$$

to calculate the cost of the electricity to heat 150 litres of water using **heater 1**.

Cost per unit of electricity = 22p

[3]

cost =

(iv) Paul has estimated that he uses 40 litres of hot water per day.
He thinks that **heater 1** would be the best one to use every day.

Explain whether you agree with Paul.

[2]

.....

.....

.....



- (b) The water in London is known to be hard.

Explain how hard water could have affected Paul's old boiler.

[2]

.....

.....

.....

- (c) Heat energy can be transferred by conduction, convection and radiation.

- (i) Explain how the shiny foam insulation used around the boiler helps to reduce heat loss. [2]

.....

.....

.....

.....

- (ii) Explain how having the foam insulation layer around the boiler benefits the environment. [2]

.....

.....

.....

.....

- (iii) Paul thinks that **heater 2** will not heat all the water in the boiler.

Use the diagram of the boiler to explain whether you agree.

[3]

.....

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.....

.....

END OF PAPER



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THE PERIODIC TABLE

Group

1 2

3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2														
7 Li Lithium 3	9 Be Beryllium 4									19 F Fluorine 9	16 O Oxygen 8	14 N Nitrogen 7	12 C Carbon 6	11 B Boron 5										
23 Na Sodium 11	24 Mg Magnesium 12									35.5 Cl Chlorine 17	32 S Sulfur 16	31 P Phosphorus 15	28 Si Silicon 14	27 Al Aluminium 13										
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36							
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	127 I Iodine 53	131 Xe Xenon 54								
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86							
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																						

Key

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number